MIL-T-45688D(MU)
31 January 1974
SUPERSEDING
MIL-T-45688C
12 December 1963

MILITARY SPECIFICATION

TELESCOPES, ARTICULATED: M105 SERIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of articulated telescope which permits the objective tube to elevate coaxially with the weapon while the eyepiece remains relatively stationary. The model designation identifies reticle pattern design and application differences (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection Of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection Of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing Of
MIL-I-45607	Inspection Equipment, Acquisition, Disposition and Maintenance Of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester, Two-Station

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F8574700	Telescope, Articulated: M105 Series
F11747310	Telescope, Articulated: M105El

PACKAGING DATA SHEET

P8574700	Packaging of Telescope, Articulated: M105 Series
P11747310	Packaging of Telescope, Articulated: M105E1

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with the drawing applicable to the type telescope specified in the contract and drawing pertinent thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Part identification and marking

3.3 Workmanship.- The requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of defects which may affect their protective qualities or appearance. Optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

3.4 Temperature conditions.- Except as specified in 3.5.1 and 3.5.2, the following requirements shall be met at a temperature between 60° and 90° Fahrenheit (F).

3.5 Environmental.

3.5.1 Storage temperatures.- The telescope shall show no evidence of physical failure such as cement separation, glass breakage or casting failure and shall meet the applicable requirements of this specification at a standard ambient temperature of 60° to 90°F after having been exposed and thermally stabilized at -80° and +160° Fahrenheit (F).

3.5.2 Extreme operating temperatures.- The telescope shall meet the applicable requirements of 3.9 while exposed and thermally stabilized at -40° and +150° F.

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3.5.3 Vibration.- The telescope shall show no evidence of physical failure due to being vibrated successively along three (3) mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either linearly or logarithmically from 5 to 55 to 5 CPS in five minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 inch (1/32 inch total excursion). Following the vibration test the telescope shall meet all subsequent requirements.

3.6 Sealing.- The telescope shall show no evidence of leakage in excess of 0.2 pounds per square inch gage when subjected to an internal pressure of 5 ± 0.10 pound per square inch (psig) for a minimum of 2 hours. The interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F.

3.7 Cleanliness.

3.7.1 Dirt.- Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any visible particle, when the reticle is brought into sharp focus, shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be four (4). The maximum number of dirt particles appearing outside this field of view shall be five (5). The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than ¼ reticle line width shall be ignored regardless of distribution.

3.7.2 Other defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations when viewing thru the objective and eye end of the instrument.

3.8 Accuracy.

3.8.1 Orientation.- The telescope shall meet the requirements of 3.8.2 to 3.8.3 inclusive while positioned as follows:

a. The geometric axis of the objective tube, as determined by the spherical surfaces of the mounting ring adjacent to Window Assembly and the mounting ring adjacent to the articulated joint, shall be in a horizontal plane.

b. The locating pin shall be set 30 degrees, plus or minus one minute counter-clockwise from the horizontal as viewed from the eyepiece.

c. The eyepiece shall form an angle of 20 ± 15′ below the geometric axis of the objective tube when measured on the hanger bracket flat surface at the eyepiece end.

3.8.2 Collimation.

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3.8.2.1 Elevation. - The line of sight through the boresight cross (zero range line) shall be positioned above the geometric axis of the telescope in accordance with the mil value indicated on the drawing for the reticle utilized, withhin plus or minus 0.5 mils.

3.8.2.2 Deflection.- The line of sight through the boresight cross of the reticle pattern utilized shall be in coincidence with a vertucal plane that passes through thr geometric a is of the objective tube within 0.50 mils.

3.8.2.3 Coincidence of reticles.- The line of sight through the boresight cross of the opposite reticle pattern shall coincide with the line of sight of thr reticle position estabished in 3.8.2.1 and 3.8.2.2 within .07 mil.

3.8.3 Parallax.- Parallax at the center of the field shall not exceed 0.05 mil when viewing a tanget within the limits of 200 to 1600 meters.

3.8.4 Parellel of reticle and image.- The vertical line of the HEAT or MIL pattern on the reticle utilized shill be barelled to the image of a verticle larger the within 30 minutes of are.

3.8.5 Reticle wander. - The image of the reticle boresight cross shall not deviate more than 1 degree from a fixed reference in an auxiliary telescope secured to the eyepiece when the eyepiece end of the telescope is elevated and depressed through its full range from stop to stop.

3.8.6 Reticle lean.- The angular rotation of the reticle image shall not exceed 1.0 degree from a fixed reference in an auxiliary telescope secured to the eyepiece when the eyepiece end of the telescope is elevated and depressed through its full range from stop to stop.

3.8.7 Image tilt.- The image of a vertical target line shall be verticle within 1 degree of arc when the telescope is positioned as in 3.8.1.

3.8.8 Resolution.- The resolution on the optical axis shall be 6.0 seconds of arc or less.

3.8.9 Eyepiece focus.- Eyepiece focus shall be adjustable over the range of plus and minus 4.0 diopters as read on the eyepiece scale. The zero diopter setting of the reticle shall indicate zero within 0.25 diopter, when the image of an infinity target is in sharp focus.

3.8.10 Reticle accuracy.- The angular value between the boresight cross and the maximum range mark for each reticle shall be as indicated on the reticle drawing within 0.3 mil.

3.9 Operability.

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3.9.1 Diopter adjustment.- The torque required to rotate the diopter adjustment shall not exceed 20 pound-inches at standard ambient temperature and 25 pound-inches at the extreme operating temperatures of 3.5.2.

3.9.2 Reticle transfer lever.- The reticle transfer lever shall be operable at the extreme operating temperatures of 3.5.2.

3.9.3 Articulated joint.- With the telescope on its side, unrestricted, the torque required to rotate the objective tube about the full excursion of the articulated joint shall not exceed 50 pound-feet at standard ambient temperature and 75 pound-feet at the extreme operating temperatures of 3.5.2.

3.10 Reticle illumination.- With light applied to the reticle illumination window the reticle marking shall be clearly defined when observed pattern is shifted into the focal plane.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the permance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirement of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval shall consist of three telescopes plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same souce of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests,the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of sQAp's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR : AQT. 0.65% Defective			
101.	Vibration	3.5.3	4.6.3
102.	Sealing	3.6	4.6.5
103.	Interior atmosphere	3.6	4.6.6
104.	Cleanliness	3.7	4.6.7
105.	Collimation, elevation	3.8.2.1	4.6.8.1
106.	Collimation, deflection	3.8.2.2	4.6.8.1
107.	Collimation, coincidence of retitles	3.8.3.2	4.6.8.2
108.	Parallax	3.8.3	4.6.8.3
109.	Parallelism of reticle and image	3.8.4	4.6.8.4
110.	Reticle wander	3.8.5	4,6.8.5
111.	Reticle lean	3.8.6	4.6.8.6
112.	Image tilt	3.8.7	4.6.8.7
113.	Resolution	3.8.8	4.6.9

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114.	Eyepiece focus	3.8.9	4.6.10
115.	Reticle accuracy	3.8.10	4.6.11
116.	Operability, diopter adjustment	3.9.1	4.6.12
117.	Operability, articulated joint	3.9.3	4.6.14
118.	Reticle illumination	3.10	4.6.15

MINOR: None Defined

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three telescopes shall be selected at random by a Government representative as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings-Visual
302.	General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three telescopes shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples The samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
303.	Storage temperature (-80°F +160°F)	3.5.1	4.6.1
304.	Operability, diopter adjustment	3.5.2, 3.9.1	4.6.2, 4.6.12
305.	Operability, reticle transfer lever	3.5.2, 3.9.2	4.6.2, 4.6.13
306.	Operability, articulated joint	3.5.2, 3.9.3	4.6.14

NOTE: The tests for Characteristics 304 through 306 inclusive shall be conducted at -40° and +150°F.

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4.4.3 Failure of sample.- Should any one item “of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures” specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within ten percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a produce which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.2 Telescope accuracy.- The testing equipment utilized to mount and orient the telescope for the accuracy tests shall permit mounting the geometric axis of the objective tube (front tube) in a horizontal plane within ten seconds: the geometric axis of the locating pin positioned 30 degrees 0.0 minutes plus or minus 10 seconds; the eyepiece housing assembly inclined below the geometric axis of the objective tube so that the angle formed by the two geometric axis (objective tube and eyepiece housing assembly) is two degrees plus or minus 15 minutes 0.0 seconds, as measured on the surface specified in 3.8.lc. The testing equipment shall also provide for articulation of the eyepiece housing assembly

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to any position in elevation between its stops. A target projector collimator having a clear aperature of at least 1.5 inch with an appropriate calibrated cross-line reticle pattern shall also be part of the testing equipment. The optical axis of the collimator projector shall be coincident to the geometric axis of the objective tube assembly of the telescope under test.

4.5.2.3.2.1 Collimation and parallax.- The testing equipment used for these tests shall be accurate within ten seconds.

4.5.2.3.2.2 Parallelism of reticle and image.- The testing equipment used for this test shall be accurate within three minutes.

4.5.2.3.2.3 Reticle wander, reticle lean, image tilt.- The testing equipment for these tests shall consist of an auxiliary telescope with a magnification of at least three power (3X). The telescope shall contain a reticle with a vertical and horizontal crossline pattern and appropriate markings for determining the limits of reticle lean and image tilt specified in 3.8.6 and 3.8.7, together with limit markings for determining wander of the articulating telescope reticle image in excess the tolerance specified in 3.8.5. The accuracy of the testing equipment for these tests shall be within six minutes.

4.5.2.3.3 Torque requirements.- A torque wrench type tester with a minimum range from zero to thirty pound-inches, the accuracy of which shall be plus or minus four ounce-inch throughout the entire zone of excursion, and a torque wrench type tester with a minimum range from zero to 100 pound-feet the accuracy of which shall be plus or minus one-half pound foot throughout the entire zone of excursion shall be utilized. The testing equipment shall be equipped with suitable adapters to accommodate testing of the diopter adjustment and objective tube.

4.5.2.3.4 Set-up and calibration.- Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the “Test Facilities” requirements of MIL-F-13926 and the conditions of 3.5.1. The telescope shall be exposed to each temperature extreme specified in 3.5.1 for a minimum of five hours. After each temperature exposure the telescope shall be stabilized in an ambient temperature between +60° and +90° Fahrenheit (F) and subjected to the visual, tactile, and audible examination of 4.6.3.1. The telescope shall conform to the requirements of 3.5.10 Caution shall be exercised-during this test to avoid subjecting the telescope to thermal shock.

4.6.1 extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the “Test Facilities” requirements

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of MIL-F-13926 and the conditions of 3.5.2. The telescope shall be exposed to each temperature extreme specified in 3.5.2 for a period of five hours. At the end of each five hour period, and while still at the temperature extreme, the telescope shall be subjected to the applicable test of 4.6.12 through 4.6.14 inclusive. The telescope shall conform to the requirements of 3.5.2. Caution shall be exercised during this test to avoid subjecting the telescope to thermal shock.

4.6.3 Vibration.- Vibrating equipment capable of providing the amplitude and frequency specified in 3.5.3 shall be utilized for this test. The vibrating equipment shall also be capable of vibrating the telescope along three mutually perpendicular axes while the telescope is maintained in a horizontal position as depicted in the lower view on sheets 1 and 2 of Drawing F8574700. The telescope shall be hortizontally mounted on the vibration table by means of an adapter. This adapter shall immobilze the objective tube assembly, and the offset tube assembly, so that the articulating joint assembly is immobilized in its midpoint of the articulating range. The telescope shall be vibrated in accordance with Procedure IX, Part I of MIL-STD-810, except that the frequency, frequency cycling, and amplitude of the vibration shall be as specified in 3.5.3. Upon completion of the vibration cycling the telescope shall be subjected to the tests specified in 4.6.4.

4.6.4 Examination.- Subsequent to vibration the telescope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.5.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts such as the articulated joint, diopter adjustment, reticle transfer lever, etc.

An audible examination shall be made of the telescope by shaking it and listening for loose internal parts or broken optical components, such as lenses, prisms, retitles, etc. Any one of the possible defects defined above shall be cause to consider a telescope defective. Telescopes which have passed the above examinations shall be subjected to the tests specified in 4.6.4 through 4.6.15 inclusive to determine overall compliance with 3.5.3.

4.6.5 Sealing.- The internal pressurizing of the telescope to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.1. The pressure shall be applied through the Valve Assembly. Pressurize the telescope with dry nitrogen gas to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure shall be retained for a minimum of 2 hours. Within the first 60 minutes of this 2 hour time period, the telescope eyepiece housing assembly shall be elevated through a minimum of 3 full excursions of the articulated joint assembly, the reticle transfer lever shall be repositioned a minimum of 3 times and the diopter adjustment shall be rotated through a minimum of 3 full excursions (cw stop-to ccw stop-to cw stop), of its rotational range. Any evidence of leakage in the telescope at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.6, as measured on the pressure gauge of the testing equipment.

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4.6.6 Interior atmosphere.- Upon satisfactory completion of the sealing test the telescope shall be flushed with the dry nitrogen gas. The flushing shall be accomplished by opening the purging vent on the objective tube and allowing the nitrogen gas to flow through the telescope, until gas bled from the telescope conforms to the dew point requirements of 3.6, as measured with a dew point tester. Close the purging vent of the telescope.

4.6.7 Cleanliness.- The telescope shall be visually examined through the eyepiece and entrance window with the unaided eye. Inspection for moisture shall be made by the technique of shadowing. Inspection for dirt or foreign particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. Cleanliness of the telescope shall conform to the requirements of 3.7.

4.6.8 Accuracy.- The following tests shall be performed with the telescope positioned in accordance with 3.8.1 on special testing equipment that conforms to the requirements of 4.5.2.3.2. The target collimator projector shall be set to a focus of 1.200 meters ± 100 meters, unless otherwise specified herein. The vertical reticle line of the target collimator shall be adjusted plumb. When the reticle pattern of the telscope under test is changed by means of the reticle transfer lever, the engagement of the reticle transfer lever with its applicable stop shall be verified prior to test.

4.6.8.1 Collimation, elevation and deflection.- The “HEP-T” and “ADPS-T” reticle pattern of the telescope shall be positioned in the field of view for this test. The angular subtense of the angle formed by projecting a line of sight through the reticle boresight point and a line of sight coincident to the intersecting point of the value specified in 3.8.2.1, as measured on the target. Any azimuth deviation of the line of sight, through the reticle boresight point with reference to the target vertical line image shall not exceed the tolerance specified in 3.8.2.2, as measured on the target. Record the postion of the applicable boresight point (intersecting point of the vertical and horizontal lines of the boresight cross) in elevation and deflection.

4.6.8.2 Collimation, coincidence of retitles.- Perform this test in sequence to the test in 4.6.8.1. Position the opposite reticle pattern “HEAT-T” in the field of view by means of the reticle positioning lever, Record the position of the reticle boresight point with reference to the target image. Any difference in the elevation or deflection position of this reticle boresight point, and the position of the reticle boresight established in 4.6.8.1 shall not exceed the tolerance specified in 3.8.2.3.

NOTE: The width (thickness) of a reticle line, in the field of view, is equal to 0.07 mil.

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4.6.8.3 Parallax.- Parallax at the optical axis (geometric center of reticle) shall not exceed the tolerance specified in 3.8.3 while viewing a target positioned at 200, 400 and 1600 meters. Parallax is recognized as any apparent movement of vertical and horizontal reticle lines with respect to the target image when the observer's head is moved from side to side or up and down.

4.6.8.4 Parallelism of reticle and image.- The "HEAT-T" reticle pattern of the telescope shall be positioned in the field of view for this test. Position one extremity of the vertical reticle line coincident to the vertical target line image. The reticle line shall be parallel to the vertical target line image within the tolerance specified in 3.8.4, as measured on the target at the opposite extremity of the reticle line. Retain the reticle pattern in the field of view for subsequent test.

4.6.8.5 Reticle wander.- Perform this test with the aid of the auxiliary telescope specified in 4.5.2.3.2.3. Affix the auxiliary telescope to the eyepiece of the telescope under test. Articulate the telescope eyepiece housing assembly to the position specified in 3.8.1.c. Center the reticle pattern of the auxiliary telescope on the image of the reticle boresight cross in the telescope under test. Elevate and depress the eyepiece housing of the telescope as required in 3.8.5 while observing the image of the reticle boresight cross at the eyepiece of the auxiliary telescope. During this articulation of the eyepiece housing the image of the reticle boresight cross shall not deviate in excess of the tolerance specified in 3.8.5, as measured by the reticle of the auxiliary telescope.

4.6.8.6 Reticle lean.- The test shall be performed with the aid of the auxiliary telescope positioned as specified in 4.6.8.5. Adjust the auxiliary telescope vertical line coincident to the image of the vertical reticle line of the telescope under test. Elevate depress the tetscope eyepiece housing as specified in 3.8.6. The image of the articulated telescope's vertical reticle line shall remain parallel to the auxiliary telescope's vertical reticle line within the tolerance specified in 3.8.6 during this articulation. Reticle lean shall be measured on tje auxiliary telescope reticle with reference to the telescope's vertical reticle line.

4.6.8.7 Image tilt.- Perform this test with the aid of an auxiliary telescope specified in 4.5.2.3.2.3. Articulate the telesope eyepice housing assembly to the position specified in 3.8.1(c). Plumb the vertical reticle line of the auxiliary telescope and position the telescope at the eyepiece of articulated telescope. Observe through the auxiliary telescope and establish one extremity of the telescope vertical reticle line coincident to the vertical target line image. The target line image shall be parallel to the auxiliary telescope vertical. reticle line within the tolerance specified in 3.8.7, as measured on the target.

4.6.9 Resolution.- Perform this test utilizing an auxiliary telescope with a magnification of at least 3 power (3X), and a resolving power chart on a wall target. The resolving power chart shall represent the angular

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subtense for the seconds of arc specified in 3.8.8, and shall contain four line sets as shown in Figure I of MIL-O-13830. The test method and procedure shall be in compliance with the "Resolution Test" specified in MIL-O-13830. The limit of resolution shall be within the value specified in 3.8.3 as determined by viewing through the auxiliary telescope.

4.6.10 Eyepiece focus.- A calibrated dioptometer shall be utilized for measuring the accuracy of the zero diopter setting of the eyepiece focus adjustment of the telescope. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter scale of the telescope to indicate zero diopter. Position the dioptometer at the telescope eyepiece. Obtain best focus of the telescope reticle at the center of the field by adjustment of the dioptometer objective scale. The telescope reticle shall be in best focus within the tolerance specified in 3.8.9, as measured on the dioptometer objective scale. Rotate the diopter scale of the telescope to its clockwise stop, note the diopter setting on the telescope diopter scale, then rotate the diopter scale to its counter clockwise stop, note diopter setting on diopter scale. The eyepiece focus adjustment range shall conform to the requirements specified in 3.8.9, as measured on the diopter scale of the telescope.

4.6.11 Reticle accuracy.- This test shall be conducted in accordance with MIL-O-13830. The angular value of the reticle pattern between the boresight cross and the maximum range mark for each reticle as measured on the target, shall not exceed the tolerance specified in 3.8.10.

4.6.12 Diopter adjustment.- Testing equipment conforming to the requirements of 4.5.2.3.3 shall be utilized for this test. The running torque required to rotate the diopter adjustment throughout its range of movement shall be within the values specified in 3.9.1 for the temperatures indicated.

4.6.13 Reticle transfer lever.- When the. telescope is thermally stabilized at -40 and +150°F the reticle transfer lever shall be subjected to a tactile inspection. The reticle transfer lever shall conform to the requirements of 3.9.2 at each extreme temperature.

4.6.14 Articulated joint.- Testing equipment conforming to the requirements of 4.5.2.3.3 shall be utilized for this test. The adapter shall be used with the torque measuring equipment to accommodate the objective tube assembly at the articulated joint. Position the telescope as specified in 3.9.3. Torque shall be measured while rotating the objective tube through its full movement about the articulated joint with the eyepiece housing assembly immobilized. The maximum running torque required to rotate the objective tube shall not exceed the value specified in 3.9.3 for the temperature incidated.

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4.6.15 Reticle illumination.- A device consisting of a .19 ampere lamp energized by 3.0 volts shall be utilized to illuminate the telescope reticle. The illuminated etchings of each reticle pattern, when positioned in the focal plane, shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the brightness of a reticle line, the defect shall

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 8574700 (M105 Series) and 11747310 (M105El). The level of protect shal be specified in the procurement document.

6. NOTES

6.1 Intended use.- The Telescope, M105 is intended for use as a component of the secondary direct fire control system of the 90mm Gun, Tank, M48 Series, the Telescopes, M105C, M105D and M105EI are intended for use as components of the secondary direct fire control system of the 105mm Gun Tank M60 and M60A1. The Telescope M105F is used for direct fire of the 165mm Gun in the Combat Engineers Vehicle, T118E1. These instructments employ a complex optical system having eight power, a 7.5° field of view, and utilize two ballistic reticles calibrated for range in meters for different type ammunition, either of which may be shifted into the focal plane. The telescopes are designed with an articulating point about midway between the objective and eyepiece to permit the objective tube of the telescope to elevate coaxially with the gun while the eyepiece movement is relatively stationary.

6.2 Ordering Data.- Procurement documents should specify the following:

Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable stock number.
d. Type of telescope to be procured (M105, M105C,M105D, M105F or M105E1).
e. Applicable packaging data sheet (see 5.1).
f. Requirement for first article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A582

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-45688D(MU) Telescopes, Articulated: M105 Series

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426

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INCH-POUND

MIL-T-45688D(AR)
AMENDMENT 2
17 June 1991
SUPERSEDING
AMENDMENT 1
20 October 1979

MILITARY SPECIFICATION

TELESCOPES, ARTICULATED: M105 SERIES

This amendment forms a part of Military Specification MIL-T-45688D(MU), dated 31 January 1974, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* 3.6, Delete and substitute:

"3.6 Sealing. The telescope shall show no evidence of leakage in excess of 0.1 pound per square inch gage when subjected to an internal pressure of 5 ± 0.10 pound per square inch (psig) for a minimum of 1 hour. The interior atmoshphere of the accepted telescope shall contain nitrogen having a dew point at least as low as - 25°F."

PAGE 4

3.8.3, Line 2, Delete "0.05 mil" and substitute ".10 mil".

PAGE 10

* 4.6.5, Delete and substitute:

"4.6.5 Sealing - The internal pressurizing of the telescope to test sealing shall be performed with a special testing equipment conforming to 4.5.2.3.1. The diopter scale shall be turned to the CW stop and remain there throughout the pressure test. The pressure shall be applied through the valve assembly. Pressurize the telescope with dry nitrogen gas to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing on sheet 1 of drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure shall be retained for a minimum of 1 hour. Within the first 30 minutes of this 1 hour time period, the telescope eyepiece housing assembly shall be elevated through a minimum of 3 full excursions of the articulated joint assembly, the reticle transfer lever shall be repositioned a minimum of 3 times. Any evidence of leakage in the telescope at the end of the 1 hour time period, shall not exceed the tolerance specified in 3.6, as measured on the pressure gauge of th testing equipment."

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NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications corrections, deletions) from the previous amendment were made. **This** was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuranceies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army - AR

Preparing activity:
Army-AR

(Project 1240-AB23)

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NOTICE OF INACTIVATION
FOR NEW DESIGN

MIL-T-45688D(AR)
NOTICE 1
28 November 1995

MILITARY SPECIFICATION

TELESCOPE, ARTICULATED: M105 SERIES

This notice should be filed in front of MIL-T-45688D (AR) dated 31 January 1974

MIL-T-45688D (AR) dated 31 January 1974, with Amendment 2, dated 17 June 1991 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity:
Army - AR

AMSC N/A

FSC 1240

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